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Codon optimization has been employed in the development of experimental nucleic acid sequences. Like viruses, which influence host cell gene expression by introducing foreign RNA or DNA into cells, such experimental nucleic acids have been explored in hopes of therapeutically influencing gene expression.

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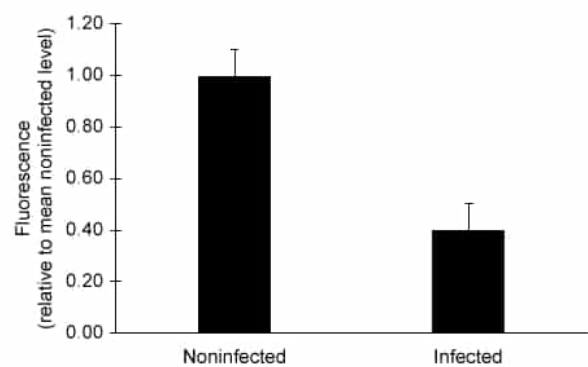


Figure 1 Effect of viral infection on splicing (Note: Bars represent means and standard errors from three experiments and are expressed relative to the mean fluorescence of noninfected cells.)

The researchers also tested the influence of codon optimization on expression of a protein encoded by an experimental messenger RNA (mRNA) therapeutic agent. Codons were optimized by replacing third-position A with G or C when possible and replacing all uridine nucleosides with N1-methyl-pseudouridine (N1-methyl-Ψ). Pseudouridine (Ψ) is a naturally occurring, post-transcriptionally modified uridine. Base pairing of selected nucleotides is shown in Table 1. Researchers concluded that codon optimization increased translational efficiency, based on a ratio of measured protein to mRNA amounts of 0.1 without optimization and 0.4 with optimization.

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Nucleotide base	Base-pairing partner(s)
A	T,U
G	C
C	G
U	A,G
Ψ	A,G,U,C
N1-methyl-Ψ	A

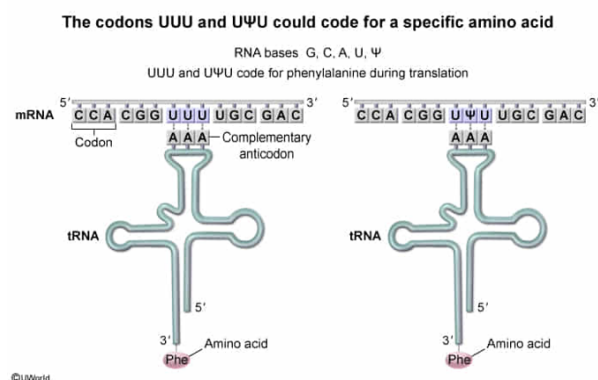
Which of the following could code for a specific amino acid during translation of the protein product of an endogenous (ie, nonviral) gene?

- ☐ A. UTU and UΨU [8%]
- ✓ ☒ B. UΨU and UUU [45%]
- ☐ C. ΨTΨ and UUU [17%]
- ☐ D. UUU and UUT [28%]

Correct

45% answered correctly

Explanation:



In humans, genomic DNA is found in nuclei and mitochondria, and RNA accumulates in the cytosol after being **transcribed** from nuclear or mitochondrial DNA. In contrast, some viruses are **composed of RNA** and thus have RNA genomes whereas other viruses employ DNA genomes like humans and other organisms. Because viruses are obligate parasites, viral genomes originate outside a host cell and migrate to specific cellular locations. Whether encoded by viral or host cell genomes, proteins are synthesized by ribosomes using mRNA as a template.

DNA and RNA are nucleic acids, composed of **nucleotides** containing various

nitrogenous bases linked to sugar molecules. Adenine, guanine, and cytosine bases are found in both DNA and RNA whereas **thymine** (T) is found **only in DNA** and **uracil** (U) is found **only in RNA**.

The passage describes how U bases can be post-transcriptionally modified to pseudouridine (Ψ), and the question asks for the nucleotide sequences that could code for a specific amino acid during translation of the protein product of an endogenous (ie, nonviral) gene. RNA contains the nucleotide U (but not T), and the passage table indicates that **Ψ can substitute for U**. Therefore, **U Ψ U and UUU** are the only two codons that **could potentially be present in RNA** and **code for a specific amino acid**.

(Choices A, C, and D) DNA, but *not* RNA, contains thymine. Therefore, the mRNA coding for the protein product of an endogenous gene will not include codons in which thymine is one of the bases.

Educational objective:

Messenger RNA is the template nucleic acid used for synthesis of proteins by ribosomes. DNA and RNA differ in base composition, with thymine present only in DNA and uracil present only in RNA.

Time spent:0 sec

Subject:
Biology

QID:403871

System:
Molecular Biology

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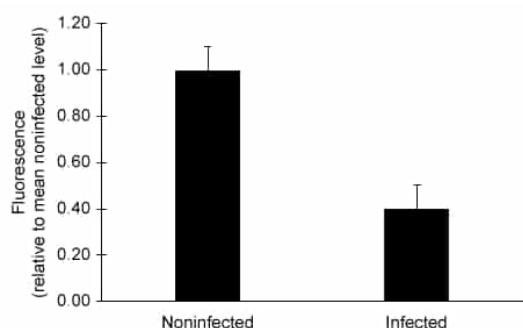


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Ψ	A,G,U,C
N1-methyl-Ψ	A

A woman is infected with a virus that uses viral RNA as a template for transcription within her body's cells. Compared to this type of viral transcription (which occurs in new membrane-bound compartments created under the direction of the virus), the chance of an endogenous gene in the infected cell being transcribed in a membrane-bound structure will most likely be:

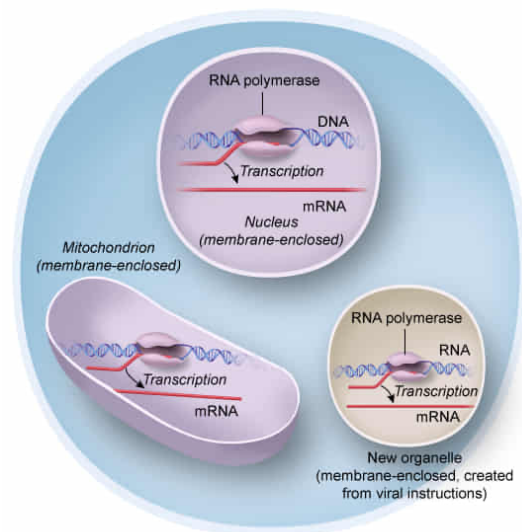
- ☐ A. higher. [22%]
- ☐ B. lower. [46%]
- ☐ C. lower, but in more than one location. [9%]
- ✓ ☒ D. approximately the same. [21%]

Correct

20% answered correctly

Explanation:

Eukaryotic transcription occurs inside organelles



According to the central dogma of molecular biology, RNA arises from DNA. The process by which this occurs is called **transcription**, which is carried out by an enzyme called RNA polymerase. During transcription, RNA polymerase uses the template strand of DNA to produce RNA with a sequence of bases closely resembling that of the coding strand of DNA.

RNA viruses have RNA genomes that, like the DNA genomes of their hosts, are used to guide the sequence in which bases are added to an RNA molecule during transcription (or to DNA during **reverse transcription**). To accomplish this task, viral RNA genomes encode a special RNA polymerase that recognizes RNA rather than DNA.

The question describes the infection of a woman with an RNA virus and states that transcription of the viral genes occurs in new membrane-bound compartments created under the direction of

the virus. The question compares the chance of an endogenous gene in an infected cell of the woman being transcribed in a membrane-bound structure against the chances for one of the viral genes to be transcribed in a membrane-bound structure.

The question states that the viral transcription occurs in membrane-bound compartments, and transcription in host cells occurs in the nucleus and in mitochondria, which are membrane-bound organelles. Therefore, the **chances of an endogenous gene being transcribed in a membrane-bound structure** is the **same as** that for a **viral gene**.

(Choices A, B, and C) These choices reflect an incorrect assessment of the chances that an endogenous gene in the infected cell will be transcribed in a membrane-bound structure. Both viral transcription and endogenous gene transcription will occur in a membrane-bound structure.

Educational objective:

Transcription is the synthesis of RNA by RNA polymerase, using the information encoded in the genome of an organism. Transcription in eukaryotes (eg, humans) occurs in the nucleus and mitochondria, which are both membrane-enclosed organelles.

Time spent:0 sec

Subject:

Biology

QID:403872

System:

Molecular Biology

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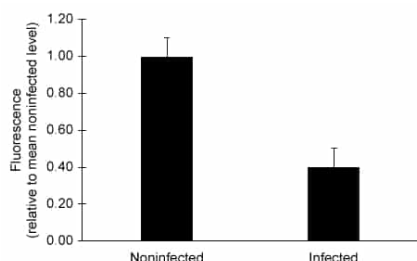


Figure 1 Effect of viral infection on splicing (Note: Bars represent means and standard errors from three experiments and are expressed relative to the mean fluorescence of noninfected cells.)

The researchers also tested the influence of codon optimization on expression of a protein encoded by an experimental messenger RNA (mRNA) therapeutic agent. Codons were optimized by replacing third-position A with G or C when possible and replacing all uridine nucleosides with N1-methyl-pseudouridine (N1-methyl-Ψ). Pseudouridine (Ψ) is a naturally occurring, post-transcriptionally modified uridine. Base pairing of selected nucleotides is shown in Table 1. Researchers concluded that codon optimization increased translational efficiency, based on a ratio of measured protein to mRNA amounts of 0.1 without optimization and 0.4 with optimization.

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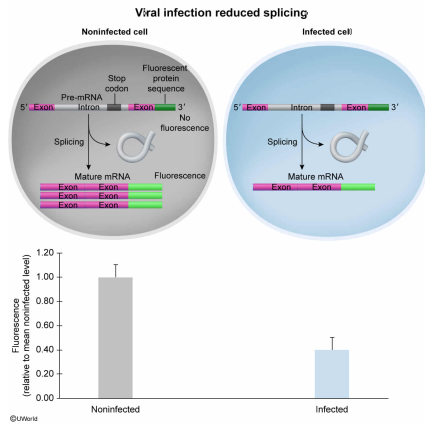
Based on the passage, infection with the virus studied by the researchers likely:

- I. increases the joining of introns separated by an exon.
 - II. decreases the joining of exons separated by an intron.
 - III. increases the number of different protein isoforms expressed.
- ☐ A. I only [11%]
☒ B. II only [63%]
☐ C. III only [14%]
☐ D. I and II only [10%]

Correct

62% answered correctly

Explanation:



The DNA sequence of a gene codes for the production of RNA, which is produced by RNA polymerase during **transcription**. Eukaryotic genes produce **RNA products** (eg, mRNA, rRNA, tRNA) that are essential for the production of proteins during translation, as well as intervening regions of RNA that are not used in rRNA, tRNA, or the production of proteins from mRNA.

To produce functional RNA products, the intervening regions of RNA (ie, introns) must be removed and the coding regions (ie, exons) must be linked together. This process, called splicing, is carried out by a complex of RNA and proteins called the **spliceosome**. In addition to linking successive exons in a gene, splicing can also join specific subsets of exons together. This process, called **alternative splicing**, can produce multiple distinct mRNA transcripts from the same gene, with the mRNA transcripts used to produce multiple distinct proteins (ie, isoforms).

The passage describes an experiment in which the effect of viral infection on RNA splicing was studied, and Figure 1 shows that splicing was reduced by infection with a specific virus. Therefore, **viral infection decreased the joining of exons separated by an intron (Number II)**.

(Number I) Introns are not joined during splicing. A reduction in splicing will have no effect on the joining of introns.

(Number III) Alternative splicing allows multiple protein isoforms to be produced from the same gene. However, Figure 1 shows that splicing was decreased by viral infection, and therefore it is likely that the number of protein isoforms expressed was reduced as well.

Educational objective:

Splicing is the process whereby the spliceosome removes noncoding intron sequences from the coding exon sequences in an RNA molecule. Alternative splicing, in which different combinations of exon sequences from the same gene are joined during splicing, can result in multiple protein isoforms produced from the same gene.

Time spent: 0 sec
Subject:
Biology

QID: 403873
System:
Molecular Biology

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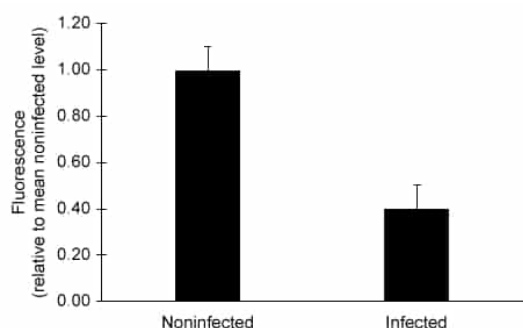


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The researchers also tested the influence of codon optimization on expression of a protein encoded by an experimental messenger RNA (mRNA) therapeutic agent. Codons were optimized by replacing third-position A with G or C when possible and replacing all uridine nucleosides with N1-methyl-pseudouridine (N1-methyl-Ψ). Pseudouridine (Ψ) is a naturally occurring, post-transcriptionally modified uridine. Base pairing of selected nucleotides is shown in Table 1. Researchers concluded that codon optimization increased translational efficiency, based on a ratio of measured protein to mRNA amounts of 0.1 without optimization and 0.4 with optimization.

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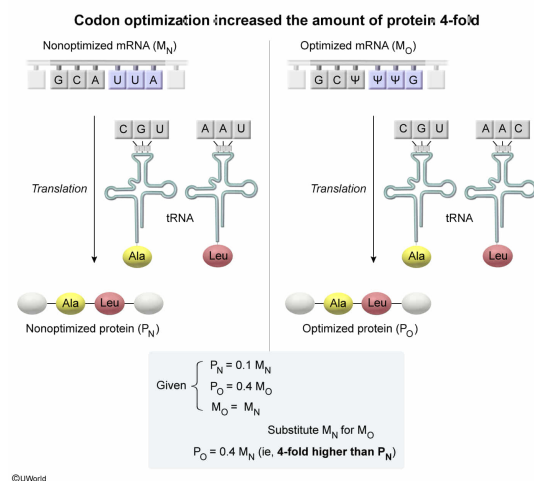
If the concentrations of the optimized and nonoptimized experimental mRNA are equal, codon optimization increases the concentration of the protein for which the codons were optimized by a factor of:

- ☐ A. 2 [9%]
☐ B. 3 [11%]
☒ C. 4 [78%]
☐ D. 5

Correct

77% answered correctly

Explanation:



During the **elongation** phase of translation, tRNA molecules deliver amino acids to the ribosome for incorporation into the growing peptide chain. Each tRNA recognizes a specific amino acid, and the identity of the next amino acid added to the polypeptide is dictated by the next codon of the mRNA translated by the ribosome. This codon is recognized by the anticodon of the appropriate tRNA, which **base-pairs** with the mRNA codon.

Although multiple codons can code for the same amino acid, some codons are preferentially utilized. As described in the passage, the presence of such "optimal" codons can increase the amount of the protein produced during translation.

The passage states that the ratio between measured protein and mRNA amounts increased from 0.1 without optimization to 0.4 with optimization. The question asks for the factor by which codon optimization increased the level of the protein studied by the researchers.

Based on the given information,

Nonoptimized protein amount = $0.1 \times$ Nonoptimized mRNA amount

Optimized protein amount = $0.4 \times$ Optimized mRNA amount

If the nonoptimized and optimized mRNA amounts are equal, then:

Optimized protein amount = $0.4 \times$ Nonoptimized mRNA amount

Because the nonoptimized protein level is 0.1 times the nonoptimized mRNA level and the optimized protein level is 0.4 times the nonoptimized mRNA, **optimization increased the protein level by a factor of 4** (ie, $0.1 \times 4 = 0.4$).

(Choices A, B, and D) These choices reflect incorrect calculations of the factor by which codon optimization increases the protein concentration.

Educational objective:

During translation, a tRNA anticodon pairs with the codon of an mRNA to ensure that the correct amino acid is incorporated into the growing polypeptide. Multiple codons can code for the same amino acid.

Time spent:0 sec

Subject:
Biology

QID:403874

System:
Molecular Biology

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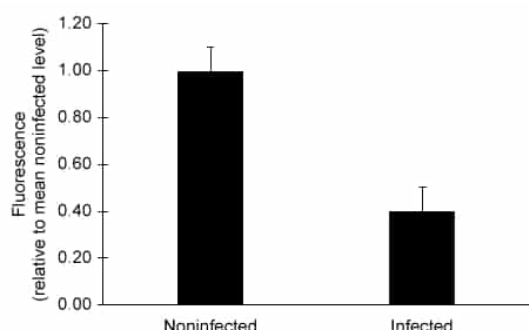


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The researchers also tested the influence of codon optimization on expression of a protein encoded by an experimental messenger RNA (mRNA) therapeutic agent. Codons were optimized by replacing third-position A with G or C when possible and replacing all uridine nucleosides with N1-methyl-pseudouridine (N1-methyl-Ψ). Pseudouridine (Ψ) is a naturally occurring, post-transcriptionally modified uridine. Base pairing of selected nucleotides is shown in Table 1. Researchers concluded that codon optimization increased translational efficiency, based on a ratio of measured protein to mRNA amounts of 0.1 without optimization and 0.4 with optimization.

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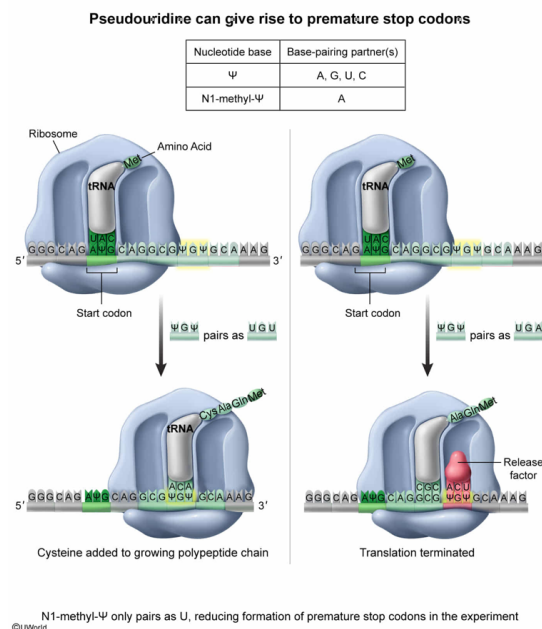
Why was mRNA modified with N1-methyl-pseudouridine (N1-methyl-Ψ) likely used in the experimental mRNA treatment instead of nonmodified mRNA?

- ☐ A. To avoid affecting methylation of endogenous molecules [12%]
- ✓ ☒ B. To reduce formation of premature stop codons [30%]
- ☐ C. To ensure anticodon pairing identical to that of U-containing codons [28%]
- ☐ D. To prevent wobble pairing of U or Ψ in the first position of stop codons in the modified mRNA [28%]

Correct

30% answered correctly

Explanation:



The sequence of amino acids in proteins is encoded by nucleotide triplets in mRNA. The four conventional RNA nucleotides are distinguished by their different bases (adenosine [A], uracil [U], guanine [G], and cytosine [C]) and provide 4³ conventional nucleotide combinations. Of these, three codons serve as stop codons (UAA, UAG, and UGA), and the single codon for methionine (AUG) also indicates the starting point for translation.

Codons serve as binding sites for tRNA molecules carrying the amino acids specified by codons or, in the case of stop codons, the release factor that terminates translation. Wobble pairing is the less-stringent pairing at the third nucleotide of a codon, allowing pairing with multiple binding partners.

The passage mentions that Ψ is present in naturally occurring mRNA and describes the incorporation of N1-methyl- Ψ into the experimental mRNA. The question asks why N1-methyl- Ψ was introduced into the experimental mRNA treatment studied by the investigators. Table 1 shows that either N1-methyl- Ψ or Ψ can substitute for uridine in mRNA, but that N1-methyl- Ψ does not participate in wobble pairing whereas Ψ is more likely than U to participate in wobble pairing.

Because of its less-stringent pairing, codons containing Ψ can pair as if they are different codons. For example, the codon **$\Psi G \Psi$ can behave like UGU**, a codon for cysteine, as well as **UGA, one of the stop codons**. Therefore, N1-methyl- Ψ is likely used in the experimental mRNA treatment instead of nonmodified mRNA to **reduce the formation of premature stop codons**.

(Choice A) The passage makes no mention of the potential influence of pseudouridine-involving methylation reactions on the methylation states of other molecules. However, if pseudouridine methylation and demethylation reactions are prevalent, then the introduction of exogenous mRNA containing *either* pseudouridine or methylated pseudouridine may affect methylation of other molecules.

(Choice C) According to Table 1, N1-methyl- Ψ is less likely to undergo wobble pairing than U. Therefore, N1-methyl- Ψ is not identical to U in this respect.

(Choice D) Although endogenous stop codons contain uridine in the first position of the codon, it is the *third* position of a codon that undergoes wobble pairing. Therefore, the U (or Ψ) in the first position of a stop codon does not participate in wobble pairing.

Educational objective:

The genetic code is said to be degenerate because multiple codons can encode the same amino acid in a protein. Wobble pairing is the less-stringent pairing of the third nucleotide of a codon.

Time spent:0 sec

Subject:
Biology

QID:403875

System:
Molecular Biology

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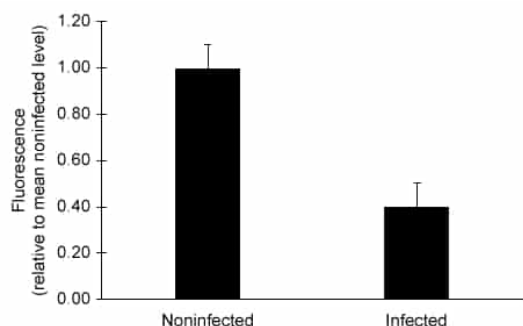


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N1-methyl-Ψ	A

Which of the following is the best experimental approach to determine whether a codon-optimized mRNA treatment affects the charging (ie, binding with an amino acid) of a specific tRNA for use in translation?

Comparison of charged and uncharged concentrations under codon-optimized and nonoptimized conditions by:

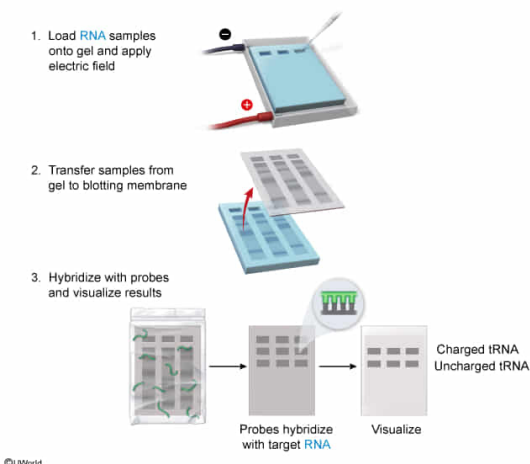
- ☐ A. absorbance measurement at 260 nm after addition of known amounts of a specific tRNA, to compare tRNA contribution to the total RNA concentration [13%]
- ☐ B. radioactive labelling to measure incorporation of ^{32}P -containing nucleotides into specific tRNA molecules [27%]
- ☐ C. western blot to measure specific tRNA molecules [14%]
- ✓ ☒ D. northern blot to measure specific tRNA molecules [44%]

Correct

43% answered correctly

Explanation:

Charged and uncharged tRNA can be measured by Northern blotting



The **central dogma** of molecular biology states that the information contained in DNA is transferred (via transcription) into RNA, and the information present in RNA is transferred into proteins (via translation). On a finer level, DNA is used as the template for the synthesis of mRNA, rRNA, and tRNA, all of which are essential for the synthesis of proteins. The **role of tRNA** in translation is to bind to specific amino acids and deliver those amino acids to the ribosome for incorporation into the growing protein.

The passage describes the use of codon optimization in the development of an experimental treatment, and the question asks for the best choice among the given options for determining

whether a codon-optimized mRNA treatment affects the charging of a specific tRNA. **Northern blotting** involves separating RNA molecules according to size using electrophoresis, followed by transfer to a special membrane ("blotting") and detection. Under appropriate conditions, **northern blotting** can be **used to separate** the **charged and uncharged** forms of a specific tRNA.

(Choice A) The absorbance of light at a wavelength of 260 nm can be used to roughly estimate total RNA (ie, mRNA, rRNA, tRNA) concentration. As such, absorbance at 260 nm would increase after the addition of known amounts of tRNA. However, neither the absorbance before nor after tRNA addition allows for the determination of the proportion of the total RNA pool that is tRNA or the extent of charging of a specific tRNA.

(Choice B) Radioactive labelling occurs during synthesis of the tRNA molecules whereas tRNA charging occurs *after* tRNA synthesis, and the incorporation of radioactive nucleotides bears no necessary relationship to tRNA charging. Therefore, the radioactivity of a specific tRNA would likely be the same, whether it was charged with an amino acid or not.

(Choice C) **Western blotting** is a method for separating and quantitating proteins (*not* tRNA).

Educational objective:

The central dogma of molecular biology is that DNA gives rise to RNA via transcription and RNA gives rise to proteins via translation. Northern blotting is a technique for separating and quantitating specific RNA molecules.

Time spent:0 sec

Subject:
Biology

QID:403876

System:
Molecular Biology